

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Basic Electrical Engg.
Examination: 1st periodical Examination
Branch: ECE, CSE, CE

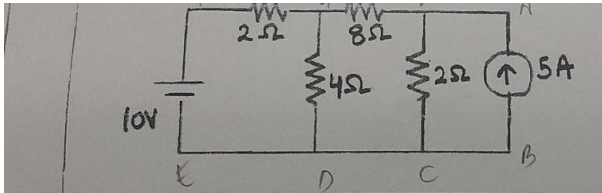
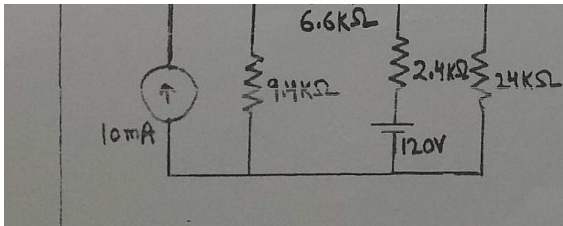
Course Code: EE-111
Scheme: NEP
Semester: 1st

Question Paper for 1st Periodical Examinations (October, 2025)

Time: One and a half hours

Max. Marks: 30

CO	Course Outcomes
CO1	Recalling Basic Electrical and magnetic laws
CO2	Remembering Steady State analysis of DC circuits
CO3	Understanding AC Single phase & Three phase Circuits
CO4	Solving complex AC, DC circuits
CO5	Explaining various concepts of Electrical machines

Q No	Questions	Marks	Targeted CO	Targeted KL
1.a	Define SuperNode	1	1	2
1.b	Compare loop and mesh?	1	1	1
1.c	Define KVL?	1	1	1
1.d	State Thevenin's Theorem?	1	1	2
1.e	State Norton's Theorem?	1	1	1
2.	Explain Maximum power transfer theorem with suitable example? What is its efficiency?	5	1	2
3.	Determine the voltage across 4 ohm resistor using Superposition Theorem? 	10	4	2
4.	Draw Norton's equivalent circuit for the figure shown and find current flowing in 24 kiloohm resistor? 	10	4	2

Course Advisor: Manavi Sharma
Assistant Professor (E.E.)